

# An Introduction to JMeter



JMeter is an Apache project. It is a load testing tool for analysing and measuring the performance of a variety of services, with a focus on Web applications.

**A**pache JMeter is an open source Apache project (available under the Apache License 2.0) that can be used for load testing. It is a pure Java application to perform load, functional, performance and regression testing. It needs JDK 5 or higher. JMeter can run on Linux, Windows and the Mac OS as a JVM.

## Installation of JMeter

- 1) JMeter's latest release can be downloaded from <http://JMeter.apache.org/>.
- 2) Click on **Download releases** in the left pane of the page.
- 3) Choose the binary file (either Zip or tgz) to download Apache JMeter.
- 4) To start JMeter, unzip the folder -> go to the bin folder -> click on the *JMeter.bat* file.

## Configuring JMeter

To make JMeter work as a browser, the following proxy settings need to be made:

- 1) Go to **Internet settings** of the browser and click on **LAN settings**.

Click on the proxy server checkbox, add 'localhost' in the address field, ensure the port is set as '8080' and click on **OK**.

## Elements of Apache JMeter

These are listed in Table 1.

The execution order of these elements is important. First, the configuration elements will execute, followed by pre-processors (if applied), then the timers (if applied), with samplers next, followed by post processors (if applied), assertions (if applied) and at the end, listeners.

Samplers and logic controllers are processed in the order in which they are available in the test plan.

## Analysis of the performance of the SUL

To analyse the performance of the application, listeners are used. Two types of functions are provided by them:

- 1) To listen/create the test results on running the script
- 2) To save, read and view the saved results

Commonly used listeners are listed below.

**View results tree:** This shows the sample response for every request; along with this, it shows the time and provides the capability to view the response received for every sample.

It provides responses in various ways such as:

- Text
- Documents
- HTML
- JSON
- XPATH tester
- CSS/JQUERY tester

But it has one disadvantage. It should not be used during load testing, as it consumes a lot of memory and